Lab Exercise 19- ClusterIP in Service in Kubernetes

A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx  # MUST match service selector
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Save file.

```
kubectl apply -f nginx-deployment.yaml
```

Verify:

```
kubectl get pods --show-labels
```

You should see:

```
app=nginx
```

```
Last login: Thu Mar 19 15:11:40 on ttys000
shivanyabharadwaj@Shivanyas-MacBook-Pro ~ % cd /Users/shivanyabharadwaj/Desktop/DevOpsLAB
shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % nano nginx-deployment.yaml

shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % kubectl apply -f nginx-deployment.yaml

deployment.apps/nginx-app created
shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % kubectl get pods --show-labels

NAME                          READY   STATUS                       RESTARTS   AGE    LABELS
nginx-app-7ccccd94f7-6rbw4    0/1     ContainerCreating            0          4s     app=nginx,pod-template
-hash=7ccccd94f7
nginx-app-7ccccd94f7-jmgfb    1/1     Running                      0          4s     app=nginx,pod-template
-hash=7ccccd94f7
pod-with-secret               0/1     CreateContainerConfigError   0          131m   <none>
test-vol1                     1/1     Running                      0          78m    <none>
test-vol2                     1/1     Running                      0          72m    <none>
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
```

```
type: ClusterIP
selector:
  app: nginx   # MUST match pod label
ports:
- port: 80
  targetPort: 80
```

Apply:

```
kubectl apply -f nginx-service.yaml
```

```
shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % nano nginx-service.yaml

shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % kubectl apply -f nginx-service.yaml

service/nginx-service created
```

Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

You should see:

```
TYPE        CLUSTER-IP
ClusterIP   10.x.x.x
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

Now you should see something like:

```
10.244.x.x:80, 10.244.x.x:80
```

```
shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % kubectl get svc

NAME            TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)   AGE
kubernetes      ClusterIP   10.96.0.1        <none>        443/TCP   3d21h
nginx-service   ClusterIP   10.105.243.208   <none>        80/TCP    6s
shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % kubectl get endpoints nginx-service

Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME            ENDPOINTS                   AGE
nginx-service   10.1.0.95:80,10.1.0.96:80   12s
```

Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

Inside pod:

```
wget -qO- http://nginx-service
```

You should see:

```
Welcome to nginx!
```

```
shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % kubectl run test-pod --image=busybox --rm -it -- sh


All commands and output from this session will be recorded in container logs, including credentials and se
nsitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
/ #
/ #
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Exit:

```
exit
```

```
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit
Session ended, resume using 'kubectl attach test-pod -c test-pod -i -t' command when the pod is running
pod "test-pod" deleted from default namespace
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

```
shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB % curl http://<CLUSTER-IP>

zsh: parse error near `\n'
shivanyabharadwaj@Shivanyas-MacBook-Pro DevOpsLAB %
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.